

# Neural\_Networks

| Task                                               | Convolutional                       | Transformer |
|----------------------------------------------------|-------------------------------------|-------------|
| Dense Prediction (segmentation & depth estimation) | FCN, U-Net, monoDepth               | DPT, SETR   |
| Image Recognition                                  | ResNet-50, EfficientNet             | ViT, DinoV2 |
| Object Detection                                   | YOLO, Faster R-CNN, CenterNet, FCOS | DETR        |
| Image Captioning                                   | CNN+LSTM,                           | CLIP        |
| Keypoint detection (local feature matching)        | Superpoint, superglue               | Loftr       |

|                               | Low dimensional states           | High-dimensional states       |
|-------------------------------|----------------------------------|-------------------------------|
| Static dataset                | Classic supervised learning (ML) | Deep supervised learning (DL) |
| Agent/environment interaction | Tabular reinforcement learning   | Deep reinforcement learning   |

## Activation functions

1. Code: mytorch/activations.py

```
import numpy as np

class Activation:
    def __init__(self):
        self.x = None
        self.update_weights = False

    def forward(self, z):
        raise NotImplementedError()

    def backward(self, dLdx):
        raise NotImplementedError()

class Identity(Activation):
    def forward(self, z):
        self.x = z
        return self.x

    def backward(self, dLdx):
        dx dz = np.ones_like(self.x)
        dLdz = dLdx * dx dz
        return dLdz

class ReLU(Activation):
    def forward(self, z):
        self.x = np.maximum(z, np.zeros_like(h))
        return self.x

    def backward(self, dLdx):
        dx dz = np.zeros_like(self.x)
        dx dz[np.where(self.x > 0)] = 1
        dLdz = dLdx * dx dz
        return dLdz

class Tanh():
    """
    Modified for RNN Classifier.
    """
    def forward(self, z):
        # self.x = (np.exp(z) - np.exp(-z)) / (np.exp(z) + np.exp(-z))
        self.x = np.tanh(z) # more stable to overflow
        return self.x

    def backward(self, dLdx, h_t=None):
        if h_t is None:
            dx dz = 1 - self.x ** 2
        else:
            dx dz = 1 - h_t ** 2
        dLdz = dLdx * dx dz
        return dLdz

class Sigmoid(Activation):
    def forward(self, z):
        self.x = 1 / (1 + np.exp(-z))
```

```

    return self.x

def backward(self, dLdx):
    dLdz = dLdx * self.x * (1 - self.x)
    return dLdz

class Softmax(Activation):
    def forward(self, z):
        exp_z = np.exp(z - np.max(z, axis = 1, keepdims=True))
        sum_exp_z = np.sum(exp_z, axis = 1, keepdims=True)
        self.x = exp_z / sum_exp_z
        return self.x

    def backward(self, dLdx):
        N = self.x.shape[0]
        C = self.x.shape[1]
        dLdz = np.zeros(shape=(N, C))
        for k in range(N):
            J = np.zeros(shape=(C, C))
            for i in range(C):
                for j in range(C):
                    if i == j:
                        J[i, j] = self.x[k, i] * (1 - self.x[k, i])
                    else:
                        J[i, j] = - self.x[k, i] * self.x[k, j]
            dLdz[k, :] = dLdx[k, :] @ J
        return dLdz

if __name__ == "__main__":
    z = np.random.randint(-5,5,(3,4))
    relu = ReLU()
    x = relu.forward(z)
    dLdx = np.random.randint(0,5,size=x.shape)
    dLdz = relu.backward(dLdx)
    print("z = ", z)
    print("x = ", x)
    print("dLdx = ", dLdx)
    print("dLdz = ", dLdz)
    print("")
    softmax = Softmax()
    x = softmax.forward(z)
    dLdz = softmax.backward(dLdx)
    print("z = ", z)
    print("x = ", x)
    print("dLdx = ", dLdx)
    print("dLdz = ", dLdz)

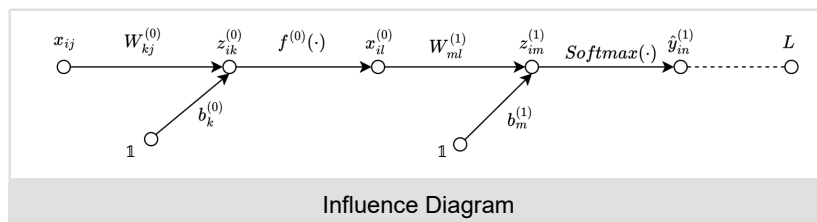
```

## Computation Graph

1. When we draw computation graph **circles represents function**, they don't represent variables. The variables are arrows between the functions, the circles are functions. So each of the circles are mathematical operations, it has inputs and outputs, and its job is to compute the derivative of its outputs wrt. to its inputs.
2. Graph structure backpropagation, also called automatic differentiation:
  1. do the following at each layer  $f(x_f) = y_f$ , starting with the last function, where  $\delta = 1$ .
  2.  $\frac{\partial L}{\partial \theta_f} = \frac{\partial f}{\partial \theta_f} \delta_{y_f}$ , local derivative  $\frac{\partial f}{\partial \theta_f}$ , upstream derivative  $\delta_{y_f}$ .

## Discriminative Models

### MLP for Classification



1. A simple MLP with a single hidden layer, i.e. one-hidden-layer neural network, with cross-entropy loss. In this model the output layer is a linear layer which will be explained later.
2. Note: the  $i$  index is specially reserved for the sample index, and that the indices  $j, k, l, m$  and  $n$  for the layer widths are all just dummy variables (no special meaning whatsoever), the 'input' is **not** a layer although historically it was referred to as the 'input layer'.

Forward variables

| Symbol    | meaning             | Shape            |                           |
|-----------|---------------------|------------------|---------------------------|
| N         | batch size          | -                |                           |
| $x$       | input               | $(N, C^{input})$ | (batch_size, input_size)  |
| $z^{(0)}$ | hidden affine layer | $(N, C^{(0)})$   | (batch_size, hidden_size) |
| $x^{(0)}$ | hidden layer        | $(N, C^{(0)})$   | (batch_size, hidden_size) |
| $z^{(1)}$ | output affine layer | $(N, C^{(1)})$   | (batch_size, output_size) |
| $y^{(1)}$ | output layer        | $(N, C^{(1)})$   | (batch_size, output_size) |

#### Model weights

| Symbol    | meaning              | Shape                    |                            |
|-----------|----------------------|--------------------------|----------------------------|
| $W^{(0)}$ | hidden layer weights | $(C^{(0)}, C^{(input)})$ | (hidden_size, input_size)  |
| $b^{(0)}$ | hidden layer bias    | $(C^{(0)}, 1)$           | (hidden_size, 1)           |
| $W^{(1)}$ | output layer weights | $(C^{(1)}, C^{(0)})$     | (output_size, hidden_size) |
| $b^{(1)}$ | output layer bias    | $(C^{(1)}, 1)$           | (output_size, 1)           |

3. Note: input\_size refers to the number of input features. It is different from the number of data samples, which in this case correspond to the batch\_size because we are using mini-batch gradient descent for training. For stochastic gradient descent the number of data samples used in each iteration to update the weights would be 1.

#### 4. Scalar form

##### 1. Forward

$$1. z_{ik}^{(0)} = \sum_j W_{kj}^{(0)} x_{ij} + b_k^{(0)}$$

$$2. x_{il}^{(0)} = f^{(0)}(z_{ik}^{(0)})$$

$$3. z_{im}^{(1)} = \sum_l W_{ml}^{(1)} x_{il}^{(0)} + b_m^{(1)}$$

$$4. y_{in}^{(1)} = \text{Softmax}(z_{im}^{(1)}) = \frac{e^{z_{in}^{(1)}}}{\sum_m e^{z_{im}^{(1)}}}$$

$$5. L = \frac{1}{N} \sum_i \sum_n y_{in} \log \frac{y_{in}}{\hat{y}_{in}^{(1)}}$$

6. where  $z$  is also called the logits, the output of the model before being normalized into probabilities.

7. The loss is a distance measure of how far the prediction score is from the ground truth score. To compare distances of two probability distributions, the Kullback-Leibler divergence is used.

##### 2. Backward

$$1. L = \frac{1}{N} \sum_i \sum_n (y_{in} \log y_{in} - y_{in} \log \hat{y}_{in})$$

2. For one-hot encoding of the ground truth probabilities the loss function will be reduced to cross-entropy loss,

$$3. L = \frac{1}{N} \sum_i \sum_n -y_{in} \log \hat{y}_{in}$$

$$4. \frac{\partial L}{\partial z_{im}^{(1)}} = \frac{1}{N} \sum_n \frac{\partial L}{\partial y_{in}} \frac{\partial y_{in}}{\partial z_{im}^{(1)}} = \frac{1}{N} (\hat{y}_{im} - y_{im})$$

3. Remark on derivative of CE Loss to the logits: To obtain  $\frac{\partial L}{\partial z_{im}^{(1)}}$ , the Jacobian for the Softmax vector activation is given by,

$$4. \frac{\partial \hat{y}_{in}}{\partial z_{im}^{(1)}} = \frac{\partial}{\partial z_{im}^{(1)}} \left( \frac{e^{z_{in}^{(1)}}}{\sum_m e^{z_{im}^{(1)}}} \right) = \begin{cases} \hat{y}_{im}(1 - \hat{y}_{im}), & \text{if } n = m \\ -\hat{y}_{im}\hat{y}_{in}, & \text{if } n \neq m \end{cases}$$

$$5. \text{ and also } \frac{\partial L}{\partial y_{in}} = -\frac{y_{in}}{\hat{y}_{in}}$$

6. Combining the above two equations,

$$\begin{aligned}
\frac{\partial L}{\partial z_{im}^{(1)}} &= \frac{1}{N} \left( \sum_{n \neq m} -\frac{y_{in}}{\hat{y}_{in}} (-\hat{y}_{im} \cancel{y_{in}}) + -\frac{y_{im}}{\hat{y}_{im}} \cancel{\hat{y}_{im}} (1 - \hat{y}_{im}) \right) \\
&= \frac{1}{N} \left( \sum_{n \neq m} y_{in} \hat{y}_{im} - y_{im} + y_{im} \hat{y}_{im} \right) \\
&= \frac{1}{N} \left( \hat{y}_{im} \underbrace{\sum_n y_{in}}_{=1 \Rightarrow \text{one-hot}} - y_{im} \right) \\
&= \frac{1}{N} (\hat{y}_{im} - y_{im})
\end{aligned}$$

nbsp;nbsp;

8.  $\frac{\partial L}{\partial x_{il}^{(0)}} = \sum_m \frac{\partial L}{\partial z_{im}^{(1)}} \frac{\partial z_{im}^{(1)}}{\partial x_{il}^{(0)}} = \sum_m \frac{\partial L}{\partial z_{im}^{(1)}} W_{ml}^{(1)}$
9.  $\frac{\partial L}{\partial W_{ml}^{(1)}} = \sum_i \frac{\partial L}{\partial z_{im}^{(1)}} \frac{\partial z_{im}^{(1)}}{\partial W_{ml}^{(1)}} = \sum_i \frac{\partial L}{\partial z_{im}^{(1)}} x_{il}^{(0)}$
10.  $\frac{\partial L}{\partial b_m^{(1)}} = \sum_i \frac{\partial L}{\partial z_{im}^{(1)}} \cdot 1$  (scalar one)
11.  $\frac{\partial L}{\partial z_{ik}^{(0)}} = \frac{\partial L}{\partial x_{il}^{(0)}} \frac{\partial x_{il}^{(0)}}{\partial z_{ik}^{(0)}}$ , for scalar activation e.g. ReLU,  $l = k$ .  $\frac{\partial L}{\partial z_{ik}^{(0)}} = \frac{\partial L}{\partial x_{ik}^{(0)}} f'(z_{ik}^{(0)})$
12.  $\frac{\partial L}{\partial x_{ij}^{(0)}} = \sum_k \frac{\partial L}{\partial z_{ik}^{(0)}} \frac{\partial z_{ik}^{(0)}}{\partial x_{ij}^{(0)}} = \sum_k \frac{\partial L}{\partial z_{ik}^{(0)}} W_{kj}^{(0)}$
13.  $\frac{\partial L}{\partial W_{kj}^{(0)}} = \sum_i \frac{\partial L}{\partial z_{ik}^{(0)}} \frac{\partial z_{ik}^{(0)}}{\partial W_{kj}^{(0)}} = \sum_i \frac{\partial L}{\partial z_{ik}^{(0)}} x_{ij}^{(0)}$
14.  $\frac{\partial L}{\partial b_k^{(0)}} \sum_i \frac{\partial L}{\partial z_{ik}^{(0)}} \frac{\partial z_{ik}^{(0)}}{\partial b_k^{(0)}} = \sum_i \frac{\partial L}{\partial z_{ik}^{(0)}} \cdot 1$  (scalar one)

## 5. Matrix form

### 1. Forward

1.  $z^{(0)} = x \cdot W^{(0)T} + b^{(0)}$
2.  $x^{(0)} = f^{(0)}(z^{(0)})$
3.  $z^{(1)} = x^{(0)} \cdot W^{(1)T} + b^{(1)}$
4.  $y_j^{(1)} = \text{Softmax}(z_j^{(1)}) = \frac{e^{z_j^{(1)}}}{\sum_j e^{z_j^{(1)}}}$
5.  $L = \frac{\iota_N(-y \odot \log \hat{y}^{(1)}) - \iota_C}{N}$

### 2. Backward

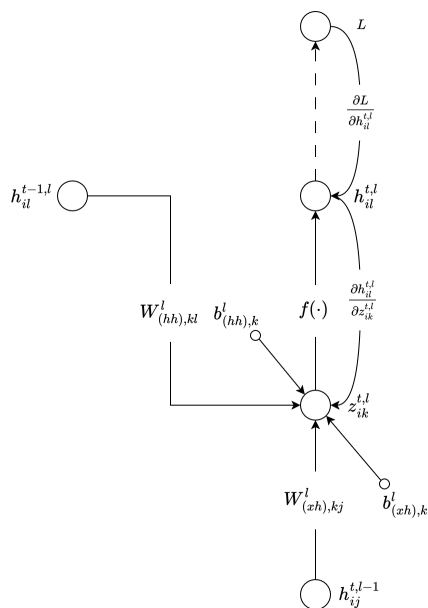
1.  $\frac{\partial L}{\partial z^{(1)}} = \frac{\hat{y}^{(1)} - y}{N}$
2.  $\frac{\partial L}{\partial x^{(0)}} = \frac{\partial L}{\partial z^{(1)}} \cdot W^{(1)}$
3.  $\frac{\partial L}{\partial W^{(1)}} = \frac{\partial L}{\partial z^{(1)}}^T x^{(0)}$
4.  $\frac{\partial L}{\partial b^{(1)}} = \frac{\partial L}{\partial z^{(1)}}^T \iota_N$
5.  $\frac{\partial L}{\partial z^{(0)}} = \frac{\partial L}{\partial x^{(0)}} \odot f'^{(0)}(z^{(0)})$
6.  $\frac{\partial L}{\partial x} = \frac{\partial L}{\partial z^{(0)}} \cdot W^{(0)}$
7.  $\frac{\partial L}{\partial W^{(0)}} = \frac{\partial L}{\partial z^{(0)}}^T x$
8.  $\frac{\partial L}{\partial b^{(0)}} = \frac{\partial L}{\partial z^{(0)}}^T \cdot \iota_N$

3. The matrix form is derived from the scalar form. The LHS and RHS of the scalar equations should conform to each other. Summation sign or lack thereof will determine the type of matrix multiplication: normal matrix multiplication  $\cdot$ , element-wise multiplication  $\odot$  or if any broadcasting is needed.

## Auto regressive Models

### RNN

1. RNN Cell (CMU 11-785)



1.

## 2. An RNN Cell forward

1. takes as input:  $x$  or  $h_{\text{prev\_l}}, h_{\text{prev\_t}}$
2. produces:  $h_{\text{next}}$  and cache:  $(h_{\text{next}}, h_{\text{prev\_l}}, h_{\text{prev\_t}}, dh_{\text{next}})$  (for backprop, in most implementations)

## 3. An RNN Cell backward

1. takes as input, upstream gradient  $dh_{\text{next}}, h_{\text{next}}, h_{\text{prev\_l}}, h_{\text{prev\_t}}$
2. produces:  $dx$  or  $dh_{\text{prev\_l}}, dh_{\text{prev\_t}}$

4. RNN is a parametric function in which itself is conditioned on its own internal state of the previous time step, and the overall effect of this RNN transforms input sequences into output sequences. In colloquial terms, RNNs have a memory of past events which is encoded in this internal state.

5. Implicit attention: The RNN will attend to some parts of this internal state more than others, i.e. selective memory, selectively looking at something and not others that it received in the past then using those to make decisions. This is why attention and memory are intimately bounded together.

## 6. Code: `mytorch/rnn_cell.py`

```
import numpy as np
from .activation import *

class RNNCell:

    def __init__(self, input_size, hidden_size):

        self.input_size = input_size
        self.hidden_size = hidden_size

        # Activation function for
        self.activation = Tanh()

        # hidden dimension and input dimension
        h = self.hidden_size
        d = self.input_size

        # Weights and biases
        self.W_ih = np.random.randn(h, d)
        self.W_hh = np.random.randn(h, h)
        self.b_ih = np.random.randn(h)
        self.b_hh = np.random.randn(h)

        self.dW_ih = np.zeros((h, d))
        self.dW_hh = np.zeros((h, h))
        self.db_ih = np.zeros(h)
        self.db_hh = np.zeros(h)

    def init_weights(self, W_ih, W_hh, b_ih, b_hh):
        self.W_ih = W_ih
        self.W_hh = W_hh
        self.b_ih = b_ih
        self.b_hh = b_hh

    def zero_grad(self):
        h, d = self.hidden_size, self.input_size
        self.dW_ih = np.zeros((h, d))
        self.dW_hh = np.zeros((h, h))
        self.db_ih = np.zeros(h)
        self.db_hh = np.zeros(h)

    def __call__(self, x, h_prev_t):
```

```

        return self.forward(x, h_prev_t)

    def forward(self, x, h_prev_t):
        h_next = h_prev_t @ self.W_hh.T + self.b_ih + x @ self.W_ih.T + self.b_hh
        out = self.activation.forward(h_next)
        return out

    def backward(self, dh_next, h_next, h_prev_l, h_prev_t):
        dz = self.activation.backward(dh_next, h_next)
        batch_size = dh_next.shape[0]
        # 1) Compute the averaged gradients of the weights and biases
        self.dW_ih += dz.T @ h_prev_l / batch_size
        self.dW_hh += dz.T @ h_prev_t / batch_size
        self.db_ih += np.sum(dz.T, axis=1) / batch_size
        self.db_hh += np.sum(dz.T, axis=1) / batch_size

        # 2) Compute dx, dh_prev_t
        dx = dz @ self.W_ih
        dh_prev_t = dz @ self.W_hh

        return dx, dh_prev_t

```

## 2. RNN Classifier (CMU 11-785)

1. Example: many-to-one, multiple layers RNN Classifier
2. RNN Classifier forward,
3. RNN Classifier backward,
  1. An important principle for BPTT: very simple rule: for each node with multiple descendants in the computational graph, simply add up the delta vectors coming from all of the descendants.
4. Code: models/rnn\_classifier.py

```

import numpy as np

from HW3.P1.mytorch.rnn_cell import *
from HW3.P1.mytorch.linear import *

class RNNPhonemeClassifier(object):
    """RNN Phoneme Classifier class."""

    def __init__(self, input_size, hidden_size, output_size, num_layers=2):
        self.input_size = input_size
        self.hidden_size = hidden_size
        self.num_layers = num_layers

        # TODO: Understand then uncomment this code :)
        self.rnn = [
            (
                RNNCell(input_size, hidden_size)
                if i == 0
                else RNNCell(hidden_size, hidden_size)
            )
            for i in range(num_layers)
        ]
        self.output_layer = Linear(hidden_size, output_size)

        # store hidden states at each time step, [(seq_len+1) * (num_layers, batch_size, hidden_size)]
        self.hiddens = []

    def init_weights(self, rnn_weights, linear_weights):
        """Initialize weights.

        Parameters
        -----
        rnn_weights:
            [
                [W_ih_l0, W_hh_l0, b_ih_l0, b_hh_l0],
                [W_ih_l1, W_hh_l1, b_ih_l1, b_hh_l1],
                ...
            ]

        linear_weights:
            [W, b]

        """
        for i, rnn_cell in enumerate(self.rnn):
            rnn_cell.init_weights(*rnn_weights[i])
        self.output_layer.W = linear_weights[0]
        self.output_layer.b = linear_weights[1].reshape(-1, 1)

    def __call__(self, x, h_0=None):
        return self.forward(x, h_0)

    def forward(self, x, h_0=None):
        """RNN forward, multiple layers, multiple time steps.
        x: input sequence: (N, T, D)
        self.x: cache (for backprop) of input sequence: (N, T, D)
        h0: initial hidden state (optional), zeros if not given: (L, N, H)

```

```

self.hiddens: cache (for backprop) of computed hidden states (h_next): (T+1, L, N, H)
hiddens: computed hidden states for all the layers (L, N, H)
h_next: hidden state (N, H)
logits: output (N, H_out)
"""

N, T, D = x.shape
L = self.num_layers
H = self.hidden_size

if h_0 is None:
    hiddens = np.zeros((L, N, H), dtype=np.float64)
else:
    L, _, H = h_0.shape
    assert L == self.num_layers, H == self.hidden_size
    hiddens = h_0
self.hiddens.append(hiddens.copy())
self.x = x
for t in range(T):
    for l in range(L):
        # Compute h_prev_l
        if l == 0:
            h_prev_l = x[:, t, :]
        else:
            h_prev_l = hiddens[l - 1, :, :]
        # Compute h_prev_t
        h_prev_t = hiddens[l]
        # Perform forward on RNN Cell
        h_next = self.rnn[l].forward(h_prev_l, h_prev_t)
        # Update hiddens for next time step
        hiddens[l] = h_next

    self.hiddens.append(hiddens.copy())

logits = self.output_layer.forward(h_next)
return logits

def backward(self, delta):
    """RNN Back Propagation Through Time (BPTT).
    N: number of samples or batch size
    L: number of layers
    H: hidden dimension
    D: input dimension
    H_out: output dimension
    delta: Upstream gradient, dL/dy from linear layer before loss function
    i.e. in last RNN Cell, h_next -> y -> L: (N, H_out)
    self.hiddens: Contains h_next computed in each RNN Cell: (T+1, L, N, H)
    self.x: input sequence: (N, T, D)
    dh_next: contains upstream gradients: (L, N, H)
    for backprop, cache = (h_next, h_prev_l, h_prev_t, dh_next)
    """

    T = self.x.shape[1]
    L = self.num_layers
    H = self.hidden_size
    N, _ = delta.shape
    dh_next = np.zeros((L, N, H), dtype=np.float64)
    # Compute upstream gradient, dL/dh_next, for last RNN Cell
    dh_next[-1] = self.output_layer.backward(delta)
    # self.output_layer.backward(delta)
    for t in reversed(range(T)):
        for l in reversed(range(L)):
            # Compute variables in cache need for backprop
            h_next = self.hiddens[t + 1][l, :, :]
            h_prev_t = self.hiddens[t][l, :, :]
            # h_prev_l can come from either input sequence or h_next from previous cell, l-1
            if l == 0:
                h_prev_l = self.x[:, t, :]
            else:
                h_prev_l = self.hiddens[t + 1][l - 1, :, :]
            # Done computing variables in cache needed for backprop
            # Perform backprop on RNN Cell
            h_prev_l, h_prev_t = self.rnn[l].backward(
                dh_next[l], h_next, h_prev_l, h_prev_t
            )
            # Update dh_next for the next previous time step
            dh_next[l] = h_prev_t
            # Add downstream gradient to the upstream gradient of the next previous cell step
            if l != 0:
                dh_next[l - 1] += h_prev_l
    return dh_next / N

if __name__ == "__main__":
    input_size = 3
    hidden_size = 4
    output_size = 5
    rnn = RNNPhonemeClassifier(input_size, hidden_size, output_size)

    N = 10
    seq_len = 3
    x = np.random.random((N, seq_len, input_size))

```

```

logits = rnn.forward(x)
# print(logits.shape)
# print(logits)
delta = np.random.random((N, output_size))
dh = rnn.backward(delta)
print(dh)
print(dh.shape)

```

|          | shape  |
|----------|--------|
| $W_{xh}$ | (D, H) |
| $W_{hh}$ | (H, H) |
| $x_t$    | (N, D) |
| $h$      | (N, H) |
| $b_h$    | (H, )  |
| $z_t$    | (N, H) |

### 3. RNN Cell (CS231n)

#### 1. forward

1.  $U = x_t \cdot W_{xh}$
2.  $V = h_{t-1} \cdot W_{hh}$
3.  $z_t = U + V + b_h$
4.  $h_t = \tanh(z_t)$
5. Code: from rnn\_layers.py

```

def rnn_step_forward(x, prev_h, Wx, Wh, b):
    """
    Run the forward pass for a single timestep of a vanilla RNN that uses a tanh
    activation function.

    The input data has dimension D, the hidden state has dimension H, and we use
    a minibatch size of N.

    Inputs:
    - x: Input data for this timestep, of shape (N, D).
    - prev_h: Hidden state from previous timestep, of shape (N, H)
    - Wx: Weight matrix for input-to-hidden connections, of shape (D, H)
    - Wh: Weight matrix for hidden-to-hidden connections, of shape (H, H)
    - b: Biases of shape (H,)

    Returns a tuple of:
    - next_h: Next hidden state, of shape (N, H)
    - cache: Tuple of values needed for the backward pass.
    """
    next_h, cache = None, None
    #####
    # TODO: Implement a single forward step for the vanilla RNN. Store the next #
    # hidden state and any values you need for the backward pass in the next_h #
    # and cache variables respectively. #
    #####
    z_t = x @ Wx + prev_h @ Wh + b
    next_h = np.tanh(z_t)
    # cache = (z_t, x, prev_h, Wx, Wh)
    cache = (next_h, x, prev_h, Wx, Wh)
    #####
    #                               END OF YOUR CODE                               #
    #####
    return next_h, cache

```

#### 2. backward

1.  $\frac{\partial L}{\partial z_t} = \frac{\partial L}{\partial h_t} \odot (1 - \tanh^2(z_t))$
2.  $\frac{\partial L}{\partial U} = \frac{\partial L}{\partial z_t}$
3.  $\frac{\partial L}{\partial V} = \frac{\partial L}{\partial z_t}$
4.  $\frac{\partial L}{\partial W_{xh}} = \left( \frac{\partial L}{\partial z_t} \cdot x_t \right)^T$
5.  $\frac{\partial L}{\partial W_{hh}} = \left( \frac{\partial L}{\partial z_t} \cdot h_{t-1} \right)^T$
6.  $\frac{\partial L}{\partial x_t} = \frac{\partial L}{\partial z_t} W_{xh}^T$
7.  $\frac{\partial L}{\partial h_{t-1}} = \frac{\partial L}{\partial z_t} W_{hh}^T$
8.  $\frac{\partial L}{\partial b} = \left( \frac{\partial L}{\partial z_t} \right) \cdot \text{sum}(\text{axis} = 0)$
9. Code: from rnn\_layers.py

```

def rnn_step_backward(dnext_h, cache):
    """
    Backward pass for a single timestep of a vanilla RNN.

```



```

Inputs:
- dnext_h: Gradient of loss with respect to next hidden state
- cache: Cache object from the forward pass

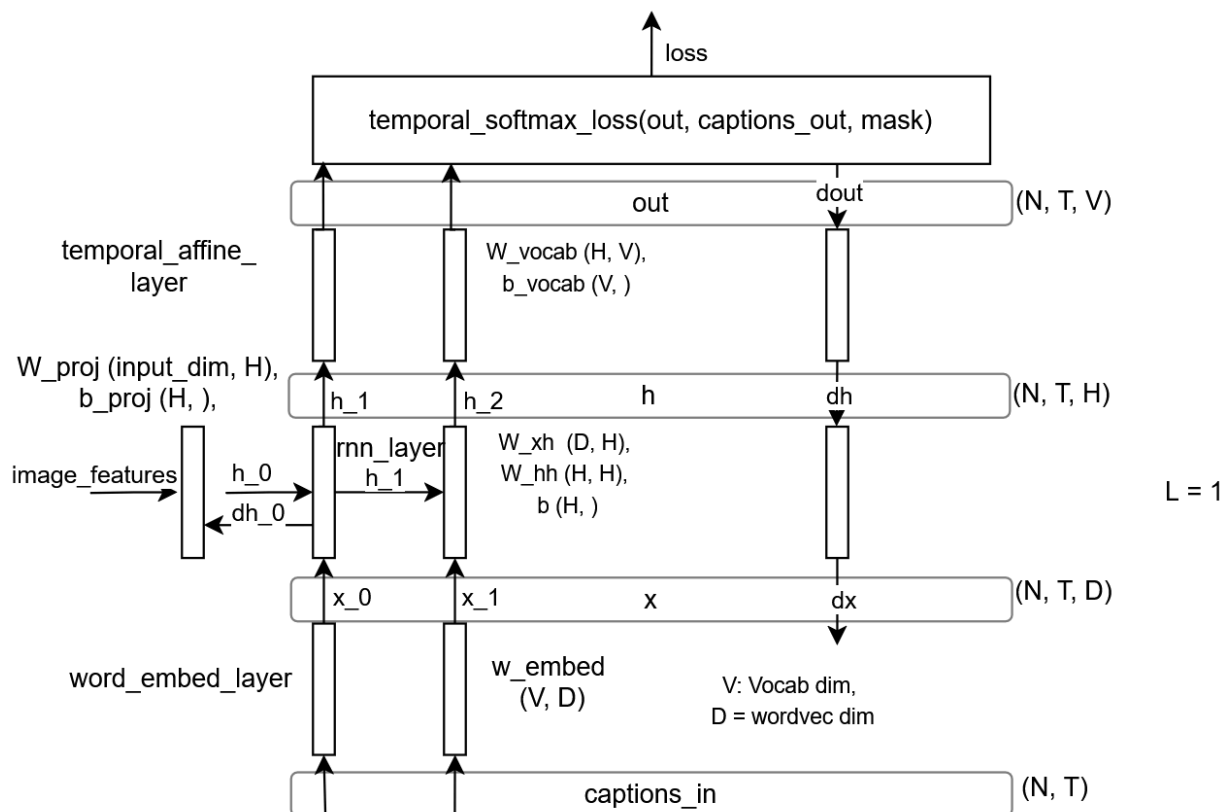
Returns a tuple of:
- dx: Gradients of input data, of shape (N, D)
- dprev_h: Gradients of previous hidden state, of shape (N, H)
- dWx: Gradients of input-to-hidden weights, of shape (D, H)
- dWh: Gradients of hidden-to-hidden weights, of shape (H, H)
- db: Gradients of bias vector, of shape (H,)
"""
dx, dprev_h, dWx, dWh, db = None, None, None, None, None
#####
# TODO: Implement the backward pass for a single step of a vanilla RNN. #
# #
# HINT: For the tanh function, you can compute the local derivative in terms #
# of the output value from tanh. #
#####
# z_t, x_t, prev_h, Wx, Wh = cache
next_h, x_t, prev_h, Wx, Wh = cache
dz = dnext_h * (1 - next_h**2)
dx = dz @ Wx.T
dprev_h = dz @ Wh.T
dWx = x_t.T @ dz
dWh = prev_h.T @ dz
db = np.sum(dz, axis=0)
#####
# #
# END OF YOUR CODE #
#####
return dx, dprev_h, dWx, dWh, db

```

#### 4. RNN Classifier (CS231n)

##### 1. many-to-many architecture for image captioning:

simple RNN Classifier for Image Captioning



##### 2. RNN Classifier forward, code given below,

##### 3. RNN Classifier backward (BPTT), code given below,

##### 4. Code: from rnn\_layers.py

```

def rnn_forward(x, h0, Wx, Wh, b):
    """
    Run a vanilla RNN forward on an entire sequence of data. We assume an input
    sequence composed of T vectors, each of dimension D. The RNN uses a hidden
    size of H, and we work over a minibatch containing N sequences. After running
    the RNN forward, we return the hidden states for all timesteps.

    Inputs:
    - x: Input data for the entire timeseries, of shape (N, T, D).
    - h0: Initial hidden state, of shape (N, H)

```

```

- Wx: Weight matrix for input-to-hidden connections, of shape (D, H)
- Wh: Weight matrix for hidden-to-hidden connections, of shape (H, H)
- b: Biases of shape (H,)

Returns a tuple of:
- h: Hidden states for the entire timeseries, of shape (N, T, H).
- cache: Values needed in the backward pass
"""
h, cache = [], []
#####
# TODO: Implement forward pass for a vanilla RNN running on a sequence of #
# input data. You should use the rnn_step_forward function that you defined #
# above. You can use a for loop to help compute the forward pass. #
#####
prev_h = h0
_, T, _ = x.shape
for t in range(T):
    next_h, next_cache = rnn_step_forward(
        x[:, t, :], prev_h,
        Wx, Wh, b
    )
    h.append(next_h)
    cache.append(next_cache)
    prev_h = next_h
h = np.array(h)
h = h.transpose(1, 0, 2)
#####
#                                     END OF YOUR CODE #
#####
return h, cache

```

```

def rnn_backward(dh, cache):
    """
    Compute the backward pass for a vanilla RNN over an entire sequence of data.

    Inputs:
    - dh: Upstream gradients of all hidden states, of shape (N, T, H)

    Returns a tuple of:
    - dx: Gradient of inputs, of shape (N, T, D)
    - dh0: Gradient of initial hidden state, of shape (N, H)
    - dWx: Gradient of input-to-hidden weights, of shape (D, H)
    - dWh: Gradient of hidden-to-hidden weights, of shape (H, H)
    - db: Gradient of biases, of shape (H,)
    """
    dx, dh0, dWx, dWh, db = None, None, None, None, None
    #####
    # TODO: Implement the backward pass for a vanilla RNN running an entire #
    # sequence of data. You should use the rnn_step_backward function that you #
    # defined above. You can use a for loop to help compute the backward pass. #
    #####
    N, T, H = dh.shape

    dx = []
    # Extract D from x.shape
    _, x, _, _ = cache[0]
    _, D = x.shape
    curr_dx = np.zeros((N, D))
    dh0 = np.zeros((N, H))
    dWx = np.zeros((D, H))
    dWh = np.zeros((H, H))
    db = np.zeros(H)
    for t in reversed(range(T)):
        # Compute upstream gradients as input to RNN cell that computes local derivatives,
        # and produces dx, dprev_h, dWh, dWx and db
        if t == T - 1:
            dnext_h = dh[:, t, :]
        else:
            dnext_h = dh[:, t, :] + dprev_h
        curr_dx, dprev_h, prev_dWx, prev_dWh, prev_db = rnn_step_backward(dnext_h, cache[t])
        dWx += prev_dWx
        dWh += prev_dWh
        db += prev_db
        dx.append(curr_dx)
    dh0 = dprev_h
    # Because the way its elements are appended, resort the order based on ascending index of time
    dx.reverse()
    dx = np.array(dx)
    dx = dx.transpose(1, 0, 2)
    #####
    #                                     END OF YOUR CODE #
    #####
    return dx, dh0, dWx, dWh, db

```

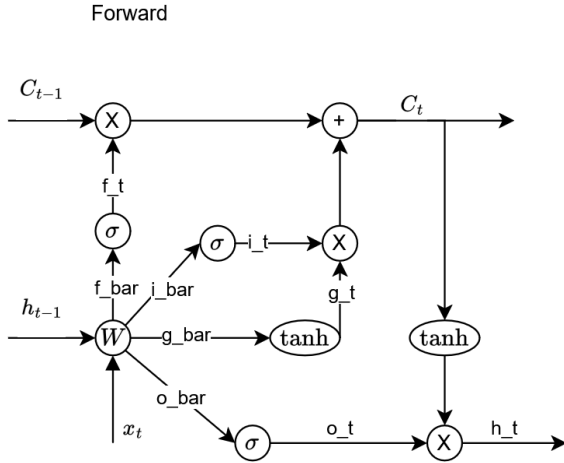
## LSTM

|           | shape     |
|-----------|-----------|
| $W_{xh}$  | (D, H)    |
| $W_{hh}$  | (H, H)    |
| $h_{t-1}$ | (N, H)    |
| $x_t$     | (N, D)    |
| $W$       | (D+H, 4H) |
| $X$       | (N, H+D)  |
| $b$       | (4H,)     |
| $z$       | (N, 4H)   |

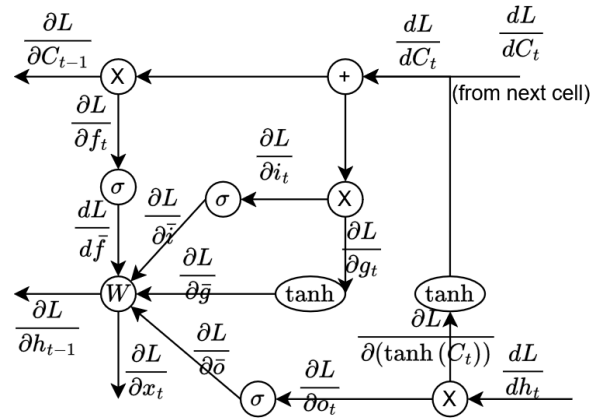
## 1. LSTM Cell

1.  $\mathbf{i}_t = \sigma(\mathbf{W}_{ix}\mathbf{x}_t + \mathbf{W}_{ih}\mathbf{h}_{t-1})$ ,
2.  $\mathbf{f}_t = \sigma(\mathbf{W}_{fx}\mathbf{x}_t + \mathbf{W}_{fh}\mathbf{h}_{t-1})$ ,
3.  $\mathbf{o}_t = \sigma(\mathbf{W}_{ox}\mathbf{x}_t + \mathbf{W}_{oh}\mathbf{h}_{t-1})$ ,
4.  $\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tanh(\mathbf{W}_{cx}\mathbf{x}_t + \mathbf{W}_{ch}\mathbf{h}_{t-1})$ ,
5.  $\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t)$ ,

LSTM Cell



Backward



## 6. Forward

1.  $i_t = \sigma([x_t, h_{t-1}] \begin{bmatrix} W_{xi} \\ W_{hi} \end{bmatrix})$ , input gate
2.  $f_t = \sigma([x_t, h_{t-1}] \begin{bmatrix} W_{xf} \\ W_{hf} \end{bmatrix})$ , forget gate
3.  $o_t = \sigma([x_t, h_{t-1}] \begin{bmatrix} W_{xo} \\ W_{ho} \end{bmatrix})$ , output gate
4.  $g_t = \tanh([x_t, h_{t-1}] \begin{bmatrix} W_{xg} \\ W_{hg} \end{bmatrix})$ , 'gate' gate
5.  $\begin{pmatrix} \bar{i} \\ \bar{f} \\ \bar{o} \\ \bar{g} \end{pmatrix} = [x_t \quad h_{t-1}] \begin{bmatrix} W_{xi} & W_{xf} & W_{xo} & W_{xg} \\ W_{hi} & W_{hf} & W_{ho} & W_{hg} \end{bmatrix} + b$ , shape: (n, 4h), where
6.  $i_t = \sigma(\bar{i})$
7.  $f_t = \sigma(\bar{f})$
8.  $o_t = \sigma(\bar{o})$
9.  $g_t = \tanh(\bar{g})$
10.  $C_t = f_t \odot C_{t-1} + i_t \odot g_t$
11.  $h_t = o_t \odot \tanh(C_t)$

## 12. Code: from rnn\_layers.py

```
def lstm_step_forward(x, prev_h, prev_c, Wx, Wh, b):
    """
    Forward pass for a single timestep of an LSTM.

    The input data has dimension D, the hidden state has dimension H, and we use
    a minibatch size of N.

    Inputs:
    - x: Input data, of shape (N, D)
    - prev_h: Previous hidden state, of shape (N, H)
    - prev_c: previous cell state, of shape (N, H)
    - Wx: Input-to-hidden weights, of shape (D, 4H)
    - Wh: Hidden-to-hidden weights, of shape (H, 4H)
    - b: Biases, of shape (4H,)

    Returns a tuple of:
    - next_h: Next hidden state, of shape (N, H)
    - next_c: Next cell state, of shape (N, H)
    - cache: Tuple of values needed for backward pass.
    """
    next_h, next_c, cache = None, None, None
    #####
    # TODO: Implement the forward pass for a single timestep of an LSTM. #
    # You may want to use the numerically stable sigmoid implementation above. #
    #####
    x_stack = np.hstack((x, prev_h))
    w_stack = np.vstack((Wx, Wh))
    z = x_stack @ w_stack + b # shape: (N, 4*H)
    i_bar, f_bar, o_bar, g_bar = np.split(z, 4, axis=1)
    i_t = sigmoid(i_bar); f_t = sigmoid(f_bar); o_t = sigmoid(o_bar); g_t = np.tanh(g_bar)
    next_c = f_t * prev_c + i_t * g_t
    next_h = o_t * np.tanh(next_c)
    cache = (prev_h, x, prev_c, next_c,
             i_bar, f_bar, o_bar, g_bar,
             i_t, f_t, o_t, g_t,
             Wx, Wh)
    #####
    #                                     END OF YOUR CODE                                     #
    #####

    return next_h, next_c, cache
```

## 7. Backward

1. cache (to store in forward):  $C_t, \bar{g}, \bar{o}, \bar{f}, \bar{i}, x_t, h_{t-1}, C_{t-1}, g_t, o_t, f_t, i_t, W_x, W_h$ .
2.  $A \cdot B$  denotes matrix multiplication,  $A \odot B$  denotes element-wise multiplication,  $\sigma(x) = \text{sigmoid}(x) = \frac{1}{1+e^{-x}}$ .
3.  $\frac{\partial L}{\partial o_t} = \frac{\partial L}{\partial h_t} \frac{\partial h_t}{\partial o_t} = \frac{\partial L}{\partial h_t} \odot \tanh(C_t)$
4.  $\frac{\partial L}{\partial (\tanh C_t)} = \frac{\partial L}{\partial h_t} \frac{\partial h_t}{\partial (\tanh C_t)} = \frac{\partial L}{\partial h_t} \odot o_t$
5.  $\frac{\partial L}{\partial C_t} + = \frac{\partial L}{\partial (\tanh(C_t))} \frac{\partial (\tanh(C_t))}{\partial C_t} = \frac{\partial L}{\partial (\tanh C_t)} \odot (1 - \tanh^2 C_t)$
6.  $\frac{\partial L}{\partial f_t} = \frac{\partial L}{\partial C_t} \odot C_{t-1}$
7.  $\frac{\partial L}{\partial C_{t-1}} = \frac{\partial L}{\partial C_t} \odot f_t$
8.  $\frac{\partial L}{\partial i_t} = \frac{\partial L}{\partial C_t} \odot g_t$
9.  $\frac{\partial L}{\partial g_t} = \frac{\partial L}{\partial C_t} \odot i_t$
10.  $\frac{\partial L}{\partial \bar{g}} = \frac{\partial L}{\partial g_t} \odot (1 - \tanh^2 \bar{g})$
11.  $\frac{\partial L}{\partial \bar{o}} = \frac{\partial L}{\partial o_t} \frac{\partial \sigma(\bar{o})}{\partial \bar{o}} = \frac{\partial L}{\partial o_t} \odot \sigma(\bar{o}) \odot (1 - \sigma(\bar{o}))$
12.  $\frac{\partial L}{\partial \bar{f}} = \frac{\partial L}{\partial f_t} \frac{\partial \sigma(\bar{f})}{\partial \bar{f}} = \frac{\partial L}{\partial f_t} \odot \sigma(\bar{f}) \odot (1 - \sigma(\bar{f}))$
13.  $\frac{\partial L}{\partial \bar{i}} = \frac{\partial L}{\partial i_t} \frac{\partial \sigma(\bar{i})}{\partial \bar{i}} = \frac{\partial L}{\partial i_t} \odot \sigma(\bar{i}) \odot (1 - \sigma(\bar{i}))$
14.  $\frac{\partial L}{\partial X} = \frac{\partial L}{\partial z} \cdot W^T$
15.  $\frac{\partial L}{\partial W} = \left( \frac{\partial L}{\partial z}^T \cdot X \right)^T$
16.  $\frac{\partial L}{\partial b} = \left( \frac{\partial L}{\partial z} \right) \cdot \text{sum}(\text{axis} = 0)$

## 17. Code: from rnn\_layers.py

```
def lstm_step_backward(dnext_h, dnext_c, cache):
    """
    Backward pass for a single timestep of an LSTM.

    Inputs:
    - dnext_h: Gradients of next hidden state, of shape (N, H)
    - dnext_c: Gradients of next cell state, of shape (N, H)
    - cache: Values from the forward pass

    Returns a tuple of:
    - dx: Gradient of input data, of shape (N, D)
    - dprev_h: Gradient of previous hidden state, of shape (N, H)
    - dprev_c: Gradient of previous cell state, of shape (N, H)
    - dWx: Gradient of input-to-hidden weights, of shape (D, 4H)
    """
```

```

- dWx: Gradient of hidden-to-hidden weights, of shape (H, 4H)
- db: Gradient of biases, of shape (4H,)
"""
dx, dh, dc, dWx, dWb, db = None, None, None, None, None, None
#####
# TODO: Implement the backward pass for a single timestep of an LSTM. #
# #
# HINT: For sigmoid and tanh you can compute local derivatives in terms of #
# the output value from the nonlinearity. #
#####
prev_h, x, prev_c, next_c, i_bar, f_bar, o_bar, g_bar, i_t, f_t, o_t, g_t, Wx, Wh = cache
do_t = dnext_h * np.tanh(next_c)
dnext_c += dnext_h * o_t * (1 - np.tanh(next_c)**2)
df_t = dnext_c * prev_c
dprev_c = dnext_c * f_t
di_t = dnext_c * g_t
dg_t = dnext_c * i_t
dg_bar = dg_t * (1 - np.tanh(g_bar)**2)
do_bar = do_t * (1 - sigmoid(o_bar)) * sigmoid(o_bar)
df_bar = df_t * (1 - sigmoid(f_bar)) * sigmoid(f_bar)
di_bar = di_t * (1 - sigmoid(i_bar)) * sigmoid(i_bar)
dz = np.hstack((di_bar, df_bar, do_bar, dg_bar))
W = np.vstack((Wx, Wh))
dx_ = dz @ W.T
_, D = x.shape
# _, H = prev_h.shape
dx = dx_[ :, :D]
dprev_h = dx_[ :, D:]
X = np.hstack((x, prev_h))
dW = (dz.T @ X).T
dWx = dW[ :D, :]
dWb = dW[D:, :]
db = dz.sum(axis=0)
#####
# END OF YOUR CODE #
#####

return dx, dprev_h, dprev_c, dWx, dWb, db

```

## 2. LSTM Classifier

### 1. Forward

#### 1. Code: from rnn\_layers.py

```

def lstm_forward(x, h0, Wx, Wh, b):
    """
    Forward pass for an LSTM over an entire sequence of data. We assume an input
    sequence composed of T vectors, each of dimension D. The LSTM uses a hidden
    size of H, and we work over a minibatch containing N sequences. After running
    the LSTM forward, we return the hidden states for all timesteps.

    Note that the initial cell state is passed as input, but the initial cell
    state is set to zero. Also note that the cell state is not returned; it is
    an internal variable to the LSTM and is not accessed from outside.

    Inputs:
    - x: Input data of shape (N, T, D)
    - h0: Initial hidden state of shape (N, H)
    - Wx: Weights for input-to-hidden connections, of shape (D, 4H)
    - Wh: Weights for hidden-to-hidden connections, of shape (H, 4H)
    - b: Biases of shape (4H,)

    Returns a tuple of:
    - h: Hidden states for all timesteps of all sequences, of shape (N, T, H)
    - cache: Values needed for the backward pass.
    """
    h, cache = [], []
    #####
    # TODO: Implement the forward pass for an LSTM over an entire timeseries. #
    # You should use the lstm_step_forward function that you just defined. #
    #####
    _, T, _ = x.shape
    prev_h = h0
    prev_c = np.zeros_like(prev_h)
    for t in range(T):
        next_h, next_c, next_cache = lstm_step_forward(x[:, t, :], prev_h, prev_c, Wx, Wh, b)
        h.append(next_h)
        cache.append(next_cache)
        prev_h = next_h
        prev_c = next_c
    h = np.array(h)
    h = h.transpose(1, 0, 2)
    #####
    # END OF YOUR CODE #
    #####

    return h, cache

```

### 2. Backward

## 1. Code: from rnn\_layers.py

```
def lstm_backward(dh, cache):
    """
    Backward pass for an LSTM over an entire sequence of data.]

    Inputs:
    - dh: Upstream gradients of hidden states, of shape (N, T, H)
    - cache: Values from the forward pass

    Returns a tuple of:
    - dx: Gradient of input data of shape (N, T, D)
    - dh0: Gradient of initial hidden state of shape (N, H)
    - dWx: Gradient of input-to-hidden weight matrix of shape (D, 4H)
    - dWh: Gradient of hidden-to-hidden weight matrix of shape (H, 4H)
    - db: Gradient of biases, of shape (4H,)
    """
    dx, dh0, dWx, dWh, db = None, None, None, None, None
    #####
    # TODO: Implement the backward pass for an LSTM over an entire timeseries. #
    # You should use the lstm_step_backward function that you just defined. #
    #####
    dx = []
    N, T, H = dh.shape
    dnext_c = np.zeros((N, H))
    x = cache[0][1]
    _, D = x.shape
    dh0 = np.zeros((N, H))
    dWx = np.zeros((D, 4 * H))
    dWh = np.zeros((H, 4 * H))
    db = np.zeros((4 * H, ))
    for t in reversed(range(T)):
        if t == T - 1:
            dnext_h = dh[:, t, :]
        else:
            dnext_h = dh[:, t, :] + dprev_h
        dprev_x, dprev_h, dprev_c, dprev_Wx, dprev_Wh, dprev_b = lstm_step_backward(dnext_h, dnext_c,
        cache[t])
        dnext_c = dprev_c
        dWx += dprev_Wx
        dWh += dprev_Wh
        db += dprev_b
        dx.append(dprev_x)
    dh0 = dprev_h
    dx.reverse()
    dx = np.array(dx)
    dx = dx.transpose(1, 0, 2)
    #####
    #                                     END OF YOUR CODE                                     #
    #####
    return dx, dh0, dWx, dWh, db
```

## Flattening

1. Flattening dimensions is often used when we are doing matrix multiplications such as when preparing input to MLP.
2. During flattening the dimensions to be flattened are 'aggregated' into one dimension, for example (3, 4, 5) into (3, 20).
3. Code:

```
# torch.flatten

x = torch.rand(3, 4, 5)
a = x.flatten(1, 2)
b = x.flatten()
print("x.shape", x.shape)
print("a.shape", a.shape)
print("b.shape", b.shape)
```

## 4. Example:

1. mfccs shape: (T, 28)
2. during training using MLP:
  1. x: frames: (N, T, D)
  2. N: batch\_size
  3. T: time\_window: 2 \* context\_window + 1
  4. D: 28 (dimension)
  5. reshape x, (N, T \* D)->mlp->phonemes (n, 42)

## Incorporating information in neural nets

1. Most of the times in neural nets we need to incorporate information using either summing or concatenation.

2. Based on information theory, as long as the information is there it should be usable. From information standpoint it does not really matter either we use summing or concatenation.
2. Sum: less memory requirement
3. Concatenate: more memory

## Labels

1. Classification labels(no spatial dimensions):  $(N, ) \rightarrow (N, \text{num\_classes})$ .  
values ranging from 0 to num\_classes-1      in one-hot in nn.CrossEntropyLoss
  2. Segmentation labels (same as classification with two spatial dimensions):  $(N, H, W) \rightarrow (N, \text{num\_classes}, H, W)$ .  
values ranging from 0 to num\_classes-1      in one-hot in nn.CrossEntropyLoss
- $\text{pred}:(N, \text{num\_classes}, H, W)$ .

## Losses

### MSE Loss

1. MSE loss:
  1. vector form:  $L_i = \frac{1}{m} \sum_i (\hat{\mathbf{x}}_i - \mathbf{x}_i)^T (\hat{\mathbf{x}}_i - \mathbf{x}_i)$ ,  $\mathbf{x}_i \in \mathbb{R}^d$ .
  2. matrix form:  $L = \sum_i \sum_j (\hat{x}_{ij} - x_{ij}) \odot (\hat{x}_{ij} - x_{ij})$ ,  $x : (N, D)$ . Notice that the square is translated into element-wise multiplication  $\odot$  in the matrix form.

### KL divergence

1. KL divergence: take the expectation of the logarithm of the ratio of  $p$  to  $q$  as if you were to assume that  $x$  is sampled through  $p$ .
2.  $KL(P||Q) = \sum_{i=1}^n p_i \log(\frac{p_i}{q_i})$ .  $p$  is the distribution we want to learn (target distribution),  $q$  is our model prediction distribution.

### Cross entropy

1. In practice,  $\text{delta} := \frac{\partial L}{\partial z}$  is computed inside the loss function e.g. CrossEntropyLoss class in classification tasks in pytorch. Also notice that delta is proportional to the error term  $y_{\text{pred}} - y_{\text{target}}$ . Because the Softmax function is already defined inside the CrossEntropyLoss, during model initialization the last layer of the model for classification is often a linear layer.

### NLL

1. Negative Log-Likelihood error is equivalent to MSE if you have unit variance  $\sigma^2 = I$ .
2. If you learn the variance, then it ends up being weighted MSE.
3. Alternatively one can use LogSoftmax layer and apply NLLoss instead of CrossEntropyLoss.

## Generative models

1. Given observed samples  $x$  from a distribution of interest, the goal of generative model is to learn to model its true distribution  $p(x)$ .
- 2.

## Diffusion models

1. Connection to VAEs
  1. Diffusion models can be considered as special case of hierarchical VAEs.
  2. However, in diffusion models:
    1. The encoder is fixed.
    2. The latent variables have the same dimension as the data.
    3. The denoising model is shared across different timestep.
    4. The model is trained with some reweighting of the variational bound.
2. DDPM is an image-generation model
  1. Generating images.
  2. Sampling images from a simple prior.
  3. Modeling the distrib of the data  $X$  of interest.

4. Computing the prob of data  $p(x)$ ,  $x$  is an image.
5. In contrast to discriminative models that models  $p(y|x)$ .
6. Training: Diffusion to get the input.
  1. Sample an image  $x$  from the training set.
  2. Sample noise  $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ .
  3. Sample a variance  $\beta \in (0, 1)$ .
  4. Get the input  $x_t = \sqrt{1 - \beta}x + \sqrt{\beta}\epsilon$ .
  5. reparametrization trick:  $x = \text{mean} + \text{std\_dev} * Z$  to sample from normal distribution.  $Z$  is drawn from a standard normal distribution.
  6. Note: reparametrization trick is a useful mental step to do backprop through a sampling process. While the gradient of an individual sample indeed does not exist or isn't meaningful, it does exist in expectation, and the expectation is all you need to backprop. Papers that refer this as a trick may come from the years 2012-2014.
7. Training: Denoising to get the output
  1. Feed the input into the network.
  2. Network: U-Net  $f_\theta$ .
  3. Predict the noise  $f_\theta(x_t, t)$ .
  4. Loss is MSE.
  5. input to model:  $x_t = \sqrt{1 - \beta}x + \sqrt{\beta}\epsilon$
  6. output:  $\tilde{\epsilon}, \tilde{x} = \frac{x_t - \sqrt{\beta}\tilde{\epsilon}}{\sqrt{1 - \beta}}$ , denoising: noisy image  $x_t$  minus the noise  $\sqrt{\beta}\epsilon$ .
8. Inference: Iterative denoising
9. DDPM does not learn the variance, just the mean of the noise  $\epsilon$ .
10. ELBO continued
  1. We need the explicit form of  $q(\mathbf{x}_t, \mathbf{x}_0)$  and  $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$  for sampling.
  2. We present the result here:  $q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1 - \bar{\alpha}_t)\mathbf{I})$ , or equivalently  $\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon_0$ , where  $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$  and  $\epsilon_0 \sim \mathcal{N}(0, \mathbf{I})$ .
  3.  $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}|\mu_q(\mathbf{x}_t, \mathbf{x}_0), \Sigma_q(t))$ , where  $\mu_q(\mathbf{x}_t, \mathbf{x}_0) = \frac{(1 - \bar{\alpha}_{t-1})\sqrt{\alpha_t}}{1 - \bar{\alpha}_t}\mathbf{x}_t + \frac{(1 - \alpha_t)\sqrt{\bar{\alpha}_{t-1}}}{1 - \bar{\alpha}_t}\mathbf{x}_0$ , and  $\Sigma_q(t) = \frac{(1 - \alpha_t)(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}\mathbf{I} = \sigma_q^2(t)\mathbf{I}$ .
  4. First way of modeling
    1. From the ELBO, we know that we have to compute the KL divergence term.  $p_\theta$  is what we can set for training, and we know that  $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$  is Gaussian, so for convenience we formulate  $p_\theta$  as a Gaussian, with the same variance as  $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$  and a learnable mean  $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}|\mu_\theta(\mathbf{x}_t, t), \sigma_q^2(t)\mathbf{I})$  where  $\mu_\theta(\mathbf{x}_t, t)$  is a neural network parametrized by  $\theta$ .
    2. Now we have  $D_{KL}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)||p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)) = \frac{1}{2\sigma_q^2(t)}\|\mu_q(\mathbf{x}_t, \mathbf{x}_0) - \mu_\theta(\mathbf{x}_t, t)\|^2$ .
    3. To match the form of mean in  $\mu_q(\mathbf{x}_t, \mathbf{x}_0) = \frac{(1 - \bar{\alpha}_{t-1})\sqrt{\alpha_t}}{1 - \bar{\alpha}_t}\mathbf{x}_t + \frac{(1 - \alpha_t)\sqrt{\bar{\alpha}_{t-1}}}{1 - \bar{\alpha}_t}\mathbf{x}_0$  we set  $\mu_\theta(\mathbf{x}_t) = \frac{(1 - \bar{\alpha}_{t-1})\sqrt{\alpha_t}}{1 - \bar{\alpha}_t}\mathbf{x}_t + \frac{(1 - \alpha_t)\sqrt{\bar{\alpha}_{t-1}}}{1 - \bar{\alpha}_t}\hat{\mathbf{x}}_\theta(\mathbf{x}_t, t)$ , where  $\hat{\mathbf{x}}_\theta(\mathbf{x}_t, t)$  is another neural network (still parametrized by  $\theta$ ) that predicts the clean image  $\mathbf{x}_0$  from the noisy image  $\mathbf{x}_t$  and time  $t$ .
  5. Second way of modeling
    1. Note that we set our neural network  $(\hat{\mathbf{x}}_\theta(\mathbf{x}_t, t))$  for predicting the image. We can actually learn to predict the noise. To see that, from  $\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon_0$  we can obtain  $\mathbf{x}_0 = \frac{1}{\sqrt{\bar{\alpha}_t}}\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}\sqrt{\alpha_t}}\epsilon_0$ .
    2.  $\mu_q(\mathbf{x}_t, \mathbf{x}_0) = \frac{1}{\sqrt{\bar{\alpha}_t}}\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}\sqrt{\alpha_t}}\epsilon_0$ .
    3. So similarly we can design our mean estimator  $\mu_\theta$  as
    4.  $\mu_\theta(\mathbf{x}_t, t) = \frac{1}{\sqrt{\bar{\alpha}_t}}\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}\sqrt{\alpha_t}}\hat{\epsilon}_\theta(\mathbf{x}_t, t)$ .

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